

An Exact-Rational Engine for Kinetic Response Functions: Architecture, Verified Results, and What It Cannot Do

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Abstract

We describe an engine that computes kinetic response functions as exact objects over the rationals $\mathbb{Q}$, without floating-point arithmetic in any derivation step, and we report what it has and has not established. The engine represents a response kernel by its exact rational Taylor coefficients, forms diagonal Padé approximants over $\mathbb{Q}$, and solves the resulting dispersion relations exactly, yielding algebraic numbers with explicit minimal polynomials. For the Landau zero-sound problem the $[4/4]$ approximant agrees with a 60-digit solution of the transcendental dispersion relation to 6.8×10^{-10} at $F_0^s = 93/10$. This revision adds four things. (i) *Real parameters.* With $F_0^s(P)$ and $m^*/m(P)$ for liquid ^3He from 0 to 29 bar (Kollar and Vollhardt’s tables, derived from Greywall’s data), the one-parameter model predicts zero sound *slower* than first sound at every pressure, which contradicts ^3He . Including $F_1^s = 3(m^*/m - 1)$ gives $c_0/c_1 = 1.036$ at 0 bar falling to 1.005 at 29 bar, and the exact Padé method extends to (F_0^s, F_1^s) with 10^{-8} agreement. (ii) *Proofs.* A Lean 4 theorem over $\mathbb{R}$ shows that every undamped root satisfies $2e^{-(2+2/F_0^s)} \leq s - 1 \leq F_0^s$, so the weak-coupling threshold law this project had only observed is now a proved lower bound, and the numerics show it is asymptotically sharp. (iii) *Neutron data.* Machine-checked kinematic identities are tested on the published IN5 dispersion of superfluid ^4He at seven pressures: the maxon crosses the roton-pair threshold 2Δ between 10 and 24 bar. (iv) *Theorem 22.6 of Villani’s lecture notes.* Its numerical worked examples are reproduced by three independent implementations, and in the case its Remark 22.8 leaves open ($d = 4, \gamma \in (-3, -2\sqrt{2})$) a heat-kernel comparison gives $m/M \geq 0.996$, hence $\bar{\gamma} \approx 3.99 > |\gamma|$. This is numerical evidence, not a proof. The negative result stands: in weak coupling no Padé approximant built at $u = 0$ places a root in $(0, 1)$, so “tracking the mode into the Landau damping continuum” is not attainable this way. A dedicated section records this and the project’s retractions.

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1 Introduction

Two distinct failure modes threaten machine-assisted theoretical physics. The first is familiar: a language model asked for a derivation may produce fluent, well-formatted mathematics that is simply wrong, and the failure is difficult to detect precisely because the output looks like the real thing. The second is older and quieter: a continuous kinetic problem discretised into `float64` loses, irrecoverably, the analytic structure that determines its qualitative behaviour. Phase mixing, Landau damping and the location of collective modes all depend on branch points and on cancellations that a floating-point pipeline reports as small residual noise.

This project responds to both by refusing floating-point arithmetic in any step that contributes to a derivation. Kernels are carried as exact rational Taylor coefficients; approximants are constructed by exact linear algebra over $\mathbb{Q}$; dispersion relations become polynomials with rational coefficients, whose roots are algebraic numbers with computable minimal polynomials. Floating point appears in exactly one place, clearly quarantined: an external validation script that checks the exact results against high-precision evaluation of the transcendental equations they are supposed to solve.

The contribution of this paper is threefold, and we separate the three deliberately.

1. **The engine** (Section 2): its architecture, and the mechanisms by which the exactness rule is enforced rather than merely documented.
2. **Verified results** (Sections 3–4): exact algebraic zero-sound velocities, a convergence study against a 60-digit reference, and a threshold law reproduced to ten digits.
3. **Contact with data and proof** (Sections 4.5, 5 and 6): real ^3He Landau parameters, real neutron-scattering data on ^4He , real-analysis theorems in Lean 4, and an independent re-examination of the numerics behind Theorem 22.6 of [1].

4. **Limits and retractions** (Section 8): what the method provably cannot reach, and which earlier claims of this project have been withdrawn. This section is not an appendix to the results; it is a result.

We state at the outset what this paper does *not* claim. The data used are published, author-processed tables: Landau parameters of ^3He derived from specific-heat measurements [7, 8], and the dispersion curve of ^4He from neutron scattering [5]. No raw instrument data is used. The Lean 4 development proves statements about *models*: a bracket on the roots of the Landau zero-sound relation, a phase-mixing identity, and kinematic identities. It does not formalise the Boltzmann equation or Theorem 22.6, and a Lean theorem here is never by itself evidence for a statement about real helium. An earlier version of this paper said no experimental data was used anywhere; that sentence is withdrawn together with the claim, recorded in Section 8, that no open dataset existed.

2 The Engine

2.1 Architecture

The engine is a three-part pipeline. Earlier versions of this repository named its stages after living scientists, which wrongly implied their involvement; the stages are now named for what they do.

LinearResponseStage produces exact rational coefficient sequences: kernels of linear response functions, and the linear density perturbation used by the nonlinear protocol.

KineticStage consumes coefficient sequences and performs the kinetic algebra: exact diagonal Padé approximation, exact convolution (Cauchy products) for nonlinear orders, exact root extraction, and symbolic kernel bounds.

AgentSocrate (orchestrator) sequences the protocols, enforces the exactness rule at stage boundaries, and writes results to a persistent store, the *Alexandrie* vault, as JSON containing exact symbolic strings rather than decimal expansions.

2.2 How the exactness rule is enforced

A documented rule that nothing checks is a comment. Three mechanisms make the “no floating point in a derivation” rule operative.

Refusal at the boundary. Entry points that accept a physical parameter reject floating-point input. Supplying F_0^s as a Python `float` raises `ScientificHonestyException` rather than silently rationalising it. This is regression-tested (`test_float_coupling_is_refused`). The rule was added after an earlier revision of this code converted F_0^s to `float` and back via `limit_denominator`, a round trip that silently perturbs the very parameter whose exactness is the point.

Exact representation end to end. Coefficients are `sympy.Rational`; approximant coefficients are obtained by solving the Padé order conditions exactly; roots are returned as exact algebraic numbers, accompanied by their minimal polynomial over $\mathbb{Q}$ when one can be computed. The stored artefacts contain strings such as `13265/11482 - 5*sqrt(4144945)/11482`, not decimals.

Identities tested against closed forms, not against tables. A hard-coded coefficient table is indistinguishable from a fabricated one. The test suite therefore rederives kernel coefficients from the closed-form generating function symbolically and compares (`test_landau_kernel_matches_closed_form`). A table that was typed in by hand, or produced by a model, fails this test unless it is right.

2.3 Export to Lean 4 and the axiom audit

Exact results are exported to a small Lean 4 development (`lean4_formalization/AgoraPhysics/Protocols.lean`) that checks the arithmetic identities underpinning each protocol. Two points of methodology matter more than the theorems themselves.

First, *grepping for the string `sorry` is not evidence of a complete proof*: a theorem can depend on `sorryAx` through an import. The development therefore ends with `#print axioms` for every theorem, so the build log carries its own evidence, and continuous integration fails if `sorryAx` appears anywhere in that output.

Beyond arithmetic identities, two files now prove real-analysis statements with `mathlib`: `ZeroSoundBracket.lean` (Section 4.4) and `PhaseMixingLink.lean` (Section 4.6). They build on lemmas from a companion project, `SocrateAI-Scientific-QuantumFluids`, which pins a different Lean version; those lemmas are restated with attribution rather than imported.

Second, the default build target is the *library*, not the executable. Building the executable forces native compilation of the entire `mathlib` dependency tree, roughly 5900 additional steps, in order to link a binary that prints a greeting; proof checking does not need it. Correcting the default target is what makes verification in CI practical at all.

3 Method: exact rational Padé extraction of collective modes

3.1 Two kernels, kept apart

Earlier revisions of this project conflated two different functions under the name “Lindhard”. They are now implemented separately and named for what they are, because the conflation produced three mutually inconsistent coefficient tables across the Python code, the Lean development and the prose.

The **Landau zero-sound kernel** is

$$\chi(s) = \frac{s}{2} \ln \frac{s+1}{s-1} - 1, \quad s = \frac{\omega}{qv_F}, \quad (1)$$

whose expansion in the variable

$$u = \frac{1}{s^2} \quad (2)$$

has purely rational coefficients

$$\chi = \sum_{k \geq 1} a_k u^k, \quad \boxed{a_k = \frac{1}{2k+1}} \quad (a_1, \dots, a_6) = \left(\frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \frac{1}{9}, \frac{1}{11}, \frac{1}{13} \right). \quad (3)$$

This identity is verified symbolically in the test suite against Eq. (1). Equation (1) is the kernel that enters the *s*-wave zero-sound dispersion relation.

A second function appears in earlier tables of this project,

$$g(x) = 1 + \frac{x^2 - 1}{2x} \ln \frac{1+x}{1-x} = \sum_{k \geq 1} \frac{2}{4k^2 - 1} x^{2k}, \quad (4)$$

with coefficients $2/3, 2/15, 2/35, \dots$. It is *not* the zero-sound kernel, it is not used for any dispersion relation here, and we make no claim identifying it with a standard response function of the Fermi liquid. It is retained, and clearly labelled, only because the sequence occurs in previously published tables of this project which readers may wish to reconcile.

3.2 Diagonal Padé approximants over $\mathbb{Q}$

Given coefficients $a_0, \dots, a_{2M}$ of $A(u) = \sum_n a_n u^n$, the diagonal $[M/M]$ approximant P/Q with $Q(0) = 1$ is fixed by

$$A(u)Q(u) - P(u) = \mathcal{O}(u^{2M+1}). \quad (5)$$

The order conditions for $n = M+1, \dots, 2M$ form an $M \times M$ rational linear system for the denominator coefficients $q_1, \dots, q_M$; the numerator then follows from the conditions $n = 0, \dots, M$. Everything is solved exactly; the engine refuses to return an approximant if the system is singular rather than falling back on a least-squares surrogate.

For $M = 1$ the result is small enough to state and to check by hand. From Eq. (3), the single order condition is

$$a_2 + q_1 a_1 = 0 \implies \frac{1}{5} + \frac{q_1}{3} = 0 \implies q_1 = -\frac{3}{5}, \quad (6)$$

so that

$$P(u) = \frac{u}{3}, \quad Q(u) = 1 - \frac{3u}{5}. \quad (7)$$

This order condition is one of the statements verified in Lean (Section 2.3); note that it has content, since it is the equation that determines q_1 rather than a restatement of its definition.

3.3 The dispersion polynomial and root admissibility

The zero-sound mode of the Landau kinetic equation with only F_0^s retained satisfies

$$\chi(s) = \frac{1}{F_0^s}. \quad (8)$$

Replacing χ by P/Q turns Eq. (8) into a polynomial equation with rational coefficients,

$$\boxed{Q(u) - F_0^s P(u) = 0} \quad (9)$$

which the engine solves exactly. An undamped mode requires phase velocity above the Fermi velocity, $s > 1$, that is a real root

$$u \in (0, 1). \quad (10)$$

Roots outside $(0, 1)$ correspond to a phase velocity inside the particle-hole continuum, where the expansion underlying Eq. (1) does not describe an undamped mode; the engine reports their absence explicitly rather than returning the nearest available number. At $M = 1$, Eqs. (7) and (9) give the closed form

$$u = \frac{15}{9 + 5F_0^s}. \quad (11)$$

4 Results

4.1 QV-01: exact zero-sound velocities

Table 1 lists exact roots returned by the engine. The ${}^3\text{He}$ -like case $F_0^s = 93/10$ is notable for being fully closed-form at $M = 1$: Eq. (11) gives $u = 15/(9 + 93/2) = 10/37$, hence

$$s = \frac{1}{\sqrt{u}} = \sqrt{\frac{37}{10}} = \frac{\sqrt{370}}{10} = 1.923538406167134\dots \quad (12)$$

At higher order the roots are algebraic numbers of degree 2, 3 and 4; the engine returns each with its minimal polynomial over $\mathbb{Q}$, which is the artefact a proof assistant can consume.

Table 1: Exact roots u of the dispersion polynomial (9) admissible in $(0, 1)$. $\text{CRootOf}(p, 0)$ denotes the first real root of p ; these are exact algebraic numbers, not numerical approximations.

F_0^s	M	exact admissible root u
93/10	1	10/37
93/10	2	$13265/11482 - 5\sqrt{4144945}/11482$
93/10	3	$\text{CRootOf}(102367u^3 - 582876u^2 + 708015u - 150150, 0)$
93/10	4	$\text{CRootOf}(2898821u^4 - 30185595u^3 + 78963885u^2 - 66591525u + 12762750, 0)$
30	1	5/53
30	2	$350/337 - \sqrt{101269}/337$
30	3	$\text{CRootOf}(6059u^3 - 32697u^2 + 34881u - 3003, 0)$
30	4	$\text{CRootOf}(172297u^4 - 1731510u^3 + 4252248u^2 - 3093090u + 255255, 0)$
1	1	none in $(0, 1)$
1	2	$1365/772 - 3\sqrt{44905}/772$

4.2 Convergence against a 60-digit reference

The exact roots are compared with a 60-digit solution of the transcendental equation (8), obtained by bracketed root-finding in `mpmath`. This comparison is the only place floating point enters, and it is external to the derivation. Table 2 reports the relative error in s .

Table 2: Relative error of the exact Padé root against the 60-digit reference solution of $\chi(s) = 1/F_0^s$. “—” marks entries where the method returns no admissible root; that outcome is discussed in Section 8 and is a property of the construction, not a failure of a particular run.

F_0^s	$M = 1$	$M = 2$	$M = 3$	$M = 4$
1/10	—	—	—	—
1/2	—	—	—	—
1	—	1.49×10^{-2}	3.87×10^{-3}	1.12×10^{-3}
93/10	2.89×10^{-3}	1.79×10^{-5}	1.11×10^{-7}	6.81×10^{-10}
30	3.20×10^{-4}	2.01×10^{-7}	1.25×10^{-10}	7.71×10^{-14}

For the strong-coupling cases the convergence is geometric and the agreement at $M = 4$ is 6.8×10^{-10} ($F_0^s = 93/10$) and 7.7×10^{-14} ($F_0^s = 30$). These two cases are asserted by the validation script: a regression that degrades them makes it exit non-zero. The intermediate case $F_0^s = 1$ is reported but not asserted, because its convergence is slow for the reason given in Section 8. Encoding the regime dependence in the test gate, rather than choosing one tolerance loose enough to pass everywhere, keeps the limitation visible.

4.3 The weak-coupling threshold law

Where the Padé construction fails, the threshold behaviour is still available analytically. As $F_0^s \rightarrow 0^+$ the mode approaches the continuum edge according to the standard asymptotic

$$s - 1 \simeq 2e^{-2-2/F_0^s}. \quad (13)$$

Table 3 compares Eq. (13) with the 60-digit root. The ratio is 1.000000 at $F_0^s = 1/20$ and $1/10$, confirming to ten digits both the asymptotic law and, independently, that Eq. (8) is the correct dispersion relation, since an error in the kernel or in the relation would not reproduce an exponential law with the correct constant.

Table 3: Weak-coupling threshold law against the 60-digit reference root.

F_0^s	$s - 1$ (reference)	$2e^{-2-2/F_0^s}$	ratio
1/20	$1.149904453 \times 10^{-18}$	$1.149904453 \times 10^{-18}$	1.0000000
1/10	$5.578936256 \times 10^{-10}$	$5.578936186 \times 10^{-10}$	1.0000000
3/20	$4.383802240 \times 10^{-7}$	$4.383771812 \times 10^{-7}$	1.0000069
3/10	$3.455570071 \times 10^{-4}$	$3.444645119 \times 10^{-4}$	1.0031716

4.4 The threshold law as a theorem

Table 3 establishes Eq. (13) numerically. It is now also a proved inequality. Write $g_3(s) = \frac{s}{2} \ln \frac{s+1}{s-1} - 1$ for the kernel (1) as a function of s . Elementary bounds on the logarithm give $\frac{1}{2} \ln \frac{2}{s-1} - 1 \leq g_3(s) \leq \frac{1}{s-1}$ for $s > 1$. Evaluating both at a root, where $g_3(s) = 1/F_0^s$, yields

$$2e^{-(2+2/F_0^s)} \leq s - 1 \leq F_0^s \quad \text{for every undamped zero-sound root.} \quad (14)$$

This is `zero_sound_bracket` in `ZeroSoundBracket.lean`. It is proved over $\mathbb{R}$ with `mathlib`, from lemmas of the companion `QuantumFluids` project, whose `zero_sound_iff` also proves that such a root *exists* for every $F_0^s > 0$. Two consequences follow.

- The lower edge of Eq. (14) is exactly the right-hand side of Eq. (13), and Table 3 shows it is attained as $F_0^s \rightarrow 0$: the proved bound is asymptotically sharp.
- Because the root exists for every $F_0^s > 0$, the Padé method’s failure at weak coupling (Section 8) is a limitation of the method, not an absence of the mode.

The validation script asserts Eq. (14) at every 60-digit reference root it computes, eight couplings from 1/20 to 30, so a violation would stop the build.

4.5 QV-01 on real ^3He parameters

The value $F_0^s = 93/10$ above is a stand-in. Real parameters are available in open form: Kollar and Vollhardt [7] tabulate, from Greywall’s specific-heat data [8], the molar volume V , the linear specific-heat coefficient γ and the compressibility κ of normal liquid ^3He at $T \rightarrow 0$ for 0–29 bar. The table is read from the text layer of their arXiv PDF, with no OCR, and checked internally:

- the $\partial\gamma/\partial P$ column against finite differences of γ , to 2×10^{-3} ;
- κ against $-V^{-1}\partial V/\partial P$, to 7×10^{-3} ;
- the prefactor of their Eq. (30) against CODATA constants, to 0.06%.

From γ one obtains m^*/m , and from their Eq. (30), $\kappa = 3.285 \times 10^{-4} \text{ bar}^{-1} (m^*/m)(1 + F_0^s)^{-1} (V/\text{cm}^3)^{5/3}$, one obtains F_0^s : 10.28 at 0 bar, rising to 73.9 at 29 bar.

On these parameters the one-parameter model fails. It predicts zero sound *slower* than first sound at every pressure (Table 4), whereas in ^3He zero sound is slightly faster. The missing ingredient is the $l = 1$ parameter $F_1^s = 3(m^*/m - 1)$, which couples the density mode to the current. With it the dispersion relation becomes

$$g_3(s) = \frac{1}{\tilde{F}(s)}, \quad \tilde{F}(s) = F_0^s + \frac{F_1^s s^2}{1 + F_1^s/3}. \quad (15)$$

Equation (15) is not taken on trust. It is checked against a direct solution of the $l = 0, 1$ moment equations of the Landau kinetic equation (a 2×2 determinant with moments evaluated by quadrature), and the two agree to working precision. First sound $c_1 = v_F \sqrt{(1 + F_0^s)(1 + F_1^s/3)}/3$ agrees with the thermodynamic $1/\sqrt{\rho\kappa}$ from the same table to 2×10^{-10} .

Table 4: Zero sound c_0 and first sound c_1 (m/s) in liquid ^3He from the Landau parameters of [7]. The one-parameter model gives $c_0 < c_1$; with F_1^s , $c_0 > c_1$ as observed.

P (bar)	F_0^s	F_1^s	c_0, F_0^s only	$c_0, (F_0^s, F_1^s)$	c_1	c_0/c_1
0	10.28	5.26	120.8	200.1	193.2	1.036
6	21.43	7.29	140.5	260.1	255.5	1.018
15	41.95	9.69	161.9	332.9	329.9	1.009
29	73.94	13.00	174.4	402.7	400.6	1.005

The exact method carries over. With $u = 1/s^2$, $a = 1 + F_1^s/3$ and $\chi \approx P/Q$, Eq. (15) becomes the polynomial $P(u) (F_0^s a u + F_1^s) - Q(u) a u = 0$ over $\mathbb{Q}$. At $F_1^s = 0$ it reduces to $u(F_0^s P - Q)$, and it reproduces $u = 10/37$ at $F_0^s = 93/10$. At the real parameters for 0 and 29 bar its $[4/4]$ root agrees with the 60-digit reference to better than 10^{-8} ; the roots lie at small u , where the approximant converges fast. One caveat applies: the absolute $c_1 = 193$ m/s at 0 bar is about 5% above commonly quoted measurements. The authors of the source table note that their compressibility is least certain below about 2 bar.

4.6 QVE-02: the nonlinear $\mathcal{O}(\epsilon^2)$ response, and what it is not

The nonlinear protocol takes $\rho^{(1)}(t) = \text{sinc } t$, forms $E^{(1)} = \int \rho^{(1)}$, and evaluates the exact Cauchy product $S^{(2)} = \rho^{(1)} E^{(1)}$ followed by one further integration. The engine returns the exact rational sequence

$$\rho^{(2)}(t) = \frac{1}{2}t^2 - \frac{1}{18}t^4 + \frac{13}{4050}t^6 - \frac{4}{33075}t^8 + \frac{73}{22325625}t^{10} + \mathcal{O}(t^{12}). \quad (16)$$

The arithmetic is exact and the convolution is correct. The sequence is, however, simply the Taylor series of

$$\frac{\text{Si}(t)^2}{2}, \quad (17)$$

which the test suite now asserts term by term. Earlier versions of this project presented Eq. (16) as the discovery of a nonlinear plasma echo. That reading is retracted: an echo is a large-time phenomenon occurring at $t = \tau k_2/(k_2 - k_1)$ and requires two pulses with distinct wavenumbers and a phase space, none of which is present here; a Taylor expansion about $t = 0$ cannot exhibit one. We retain the protocol as what it is, an exact second-order Volterra response, and test it against its closed form so the identification cannot be lost again.

The input has a precise physical meaning, now proved. For free transport $\partial_t f + v \partial_x f = 0$ the k -th density mode is $\int e^{-ikv} g(v) dv$, and for the flat-top distribution $g = \frac{1}{2} \mathbb{1}_{[-1,1]}$ it equals $\sin(kt)/(kt)$ (`flat_top_mode` in `PhaseMixingLink.lean`), which decays to zero like $1/t$ (`flat_top_mode_tendsto_zero`). By Archimedes' hat-box theorem the flat top is the one-dimensional marginal of the uniform measure on a Fermi surface S^2 , so $\rho^{(1)}$ is also that surface's free-streaming mode; the test suite checks this numerically. The decay is algebraic phase mixing, in contrast with the Gaussian $e^{-k^2 t^2/2}$ of a Maxwellian. The second-order term tends to $\text{Si}(\infty)^2/2 = \pi^2/8$: it approaches a constant, and nothing echoes.

4.7 Q-RHK-02: symbolic kernel bounds

For the phenomenological angular kernel

$$\beta(\cos \theta) = \frac{1}{2} (1 + \cos^2 \theta) \exp\left[-\frac{1}{10} (1 - \cos \theta)\right], \quad (18)$$

the engine computes exactly

$$M_r = 1, \quad m_r = \left[\frac{1}{2} \left(-10 + 3\sqrt{11} \right)^2 + \frac{1}{2} \right] e^{-11/10 + 3\sqrt{11}/10}, \quad (19)$$

$$\Sigma(\beta) = \frac{\pi (910 - 1110 e^{-1/5})}{2} = 1.8988792618620465 \dots, \quad (20)$$

$$\gamma_{\text{bound}} = \frac{m_r}{M_r} + \frac{3}{2} = 1.9512876598772344 \dots \quad (21)$$

Three caveats are essential. First, β is an analytic model chosen for its forward-peaked shape; no measured data enters, and nothing here validates it against ^4He . Second, the combination in Eq. (21) was previously attributed in this repository to “Theorem 22.6” of Villani [1]; *that attribution has been checked against the source and found mismatched*: Villani’s Theorem 22.6 is real and does bound a ratio m_r/M_r , but its actual conclusion is the multiplicative, dimension-dependent bound $2\sqrt{d} \cdot \sqrt{m_r/M_r}$, used to show that Fisher information is nonincreasing along the spatially homogeneous Boltzmann equation — not the additive form of Eq. (21), which is an independently-defined convention of this project. The method has accordingly stayed named `kernel_regularity_bounds` (renamed from `apply_theorem_22_6`) so that the interface does not assert a citation the source does not support. Third, the Lean development carries the rational surrogate $16063/8232 = 1.951287657920311 \dots$, which differs from Eq. (21) by about 2.0×10^{-9} : what is checked there is arithmetic about that rational, not about the algebraic number.

What Theorem 22.6 does give for this kernel can be computed from the theorem as stated. For $B = |v - v_*|^\gamma \beta(\cos \theta)$ in $d = 3$ its radial condition reads $|\gamma| \leq 2\sqrt{3m/M}$, where $m \leq \beta/\beta_0 \leq M$ for some mixture β_0 of heat kernels on S^2 . A linear program over finite mixtures finds $m/M = 0.99654$. The symmetrised kernel is close to $\frac{4}{3} + \frac{2}{3}P_2(\cos \theta)$, which a single heat kernel matches at $t = \ln(10)/6 \approx 0.384$; the optimiser found 0.3846. Hence $\bar{\gamma} = 3.458$, close to the ceiling $2\sqrt{3}$. This is the quantity the retracted formula was meant to be, and it is a property of the model kernel only (Figure 3).

4.8 Q-RIP-03: a definitional identity, reported as such

The remaining protocol evaluates $L_* = 2d$ at $d = 2$ and obtains 4. With L_* defined as $2d$, this establishes that $2 \cdot 2 = 4$ and nothing else. It is not evidence that any Bakry–Émery curvature-dimension constant equals 4, and the earlier claim that it “protects 2D quantum films from Landau damping collapse” has no support in this work. An earlier implementation constructed a `sympy.diffgeom` metric and Ricci tensor and then discarded them, returning $2 \times \text{dim}$; that decorative machinery has been removed and the quantity is labelled a conjecture in both the code and the stored artefacts.

5 The numerics behind Theorem 22.6

Theorem 22.6 of Villani’s lecture notes [1] gives a sufficient condition for the Fisher information to be nonincreasing along the spatially homogeneous Boltzmann equation. The condition compares the angular kernel with a mixture of heat kernels, $m \leq \beta/\beta_0 \leq M$, and in its literal form reads $\sup_\theta \frac{r}{B} |\partial_r B| \leq 2\sqrt{dm/M}$. For power laws, $B = |v - v_*|^\gamma \beta(\cos \theta)$, this becomes $|\gamma| \leq \bar{\gamma} = 2\sqrt{dm/M}$. The worked examples after the theorem were computed numerically by Silvestre, published in [3], with code in [4], and the notes describe the procedure as “not a purely mathematical proof”. We re-examined them.

Reproduction, three ways. The authors’ Julia code was re-executed at a pinned commit. Because that repository carries no licence, it is run but never copied. Two further implementations, in Python and in Rust, were written from the mathematics rather than translated:

- the collision kernel comes from classical scattering, parametrised by $\beta = p^2/r_0^2$, with the deflection derivative taken under the integral; it reproduces Rutherford’s cross-section to 10^{-12} in $d = 2, 3, 4$;
- the subordinated kernel comes from the spherical heat-kernel expansion, each time integral being an exact incomplete gamma function;
- both kernels are normalised by exact $\theta \rightarrow 0$ constants.

Both quoted bounds reproduce: the minimum over the tabulated ν is 4.355 in $d = 3$ (claim ≥ 4.3) and 3.364 in $d = 2$ (claim > 3.3). The kernels agree with the authors’ code to 2×10^{-6} (collision) and 0.7% (subordinated; theirs is bump-averaged), and Rust agrees with Python to 2×10^{-9} .

One detail differs. The authors’ adaptive quadrature of the integrals defining Λ_b underestimates it as $\nu \rightarrow 2$, where the integrand tends to t^{-1} : at $d = 2$, $\nu = 1.999$ it gives 3.932 against the exact 3.997. The exact value comes from the closed form $\int_0^\infty t^{-1-s}(e^{-\alpha t} - e^{-\beta t}) dt = \Gamma(-s)(\alpha^s - \beta^s)$, confirmed to twelve digits by a singularity-resolving substitution. The error is conservative: the tabulated bounds are slightly low.

An exact identity anchors the comparison. At $d = 2$, $\nu = 1$ (two-dimensional Coulomb scattering) the symmetrised Rutherford kernel $1/\sin^2 \theta$ equals the symmetrised kernel of $(-\Delta)^{1/2}$ on the circle. The two independent computations agree to 6×10^{-11} , and the bound there is exactly $2 \cdot 2^{3/4}$.

Denser sampling. The notes sample the kernel ratio at eleven angles and 8–10 values of ν . Sampling can only overestimate an infimum, so we used 52 values of ν and up to 400 angles (Figure 1). Both claims survive, and the eleven-angle grid overestimates the bound by at most 0.021. In $d = 2$ the dense sweep finds a dip between the tabulated $\nu = 1.0$ and 1.25: the minimum is 3.3593 at $\nu \approx 1.06$, still above 3.3.

The weights. A grid search over the two-parameter families that contain the notes’ “trial and error” weights, $1 \mp a(1 - e^{-bt})$, finds them within 0.06 of the best in their family (Figure 2). Without tuning, using the plain fractional Laplacian, both claims fail over most of their range.

The case Remark 22.8 leaves open. The notes state that $\gamma \in (-3, -2\sqrt{2}]$ in $d = 4$ is not covered. For a force $\propto r^{-s}$ in dimension d , classical scattering gives $\nu = (d-1)/(s-1)$ and $\gamma = 1-2\nu$, so the open range is $\nu \in [1.9142, 2)$. The literal criterion then requires $m/M > ((2\nu-1)/4)^2$, at most 9/16. That form of the criterion reproduces three of the notes’ own statements: $\sqrt{7.5}$, $\sqrt{12/1.1}$ and $2 \cdot 2^{1/4}$. A linear program maximises m/M over nonnegative combinations of three families of comparison kernels, each admissible for $\nu < 2$:

- tempered singular kernels, with weights $t^{-1-\nu/2}e^{-\mu t}$;
- bounded kernels, with weights $t^{-1-\nu/2}(1 - e^{-bt})$;
- pure heat kernels on S^3 .

It finds $m/M \geq 0.996$ at all sixteen sampled ν , re-checked on 800 angles down to $\theta = 0.01$, hence $\bar{\gamma} \approx 3.99 > |\gamma|$. The result survives halving the basis (0.9954) and using the tempered family alone (0.9928). It fails without tempering: m/M drops to 0.48, below the required 9/16.

This is numerical evidence at the standard of the notes’ own worked examples, not a proof. It assumes Theorem 22.6 holds as stated in $d = 4$; below $\theta = 0.01$ it relies on the exact $\theta \rightarrow 0$ limit; and it is offered for the authors to check. A numerical artefact met along the way is documented in the repository: a single spectral cut-off chosen for the smallest angle loses accuracy near $\pi/2$ in $d = 4$, and is fixed by per-block cut-offs.

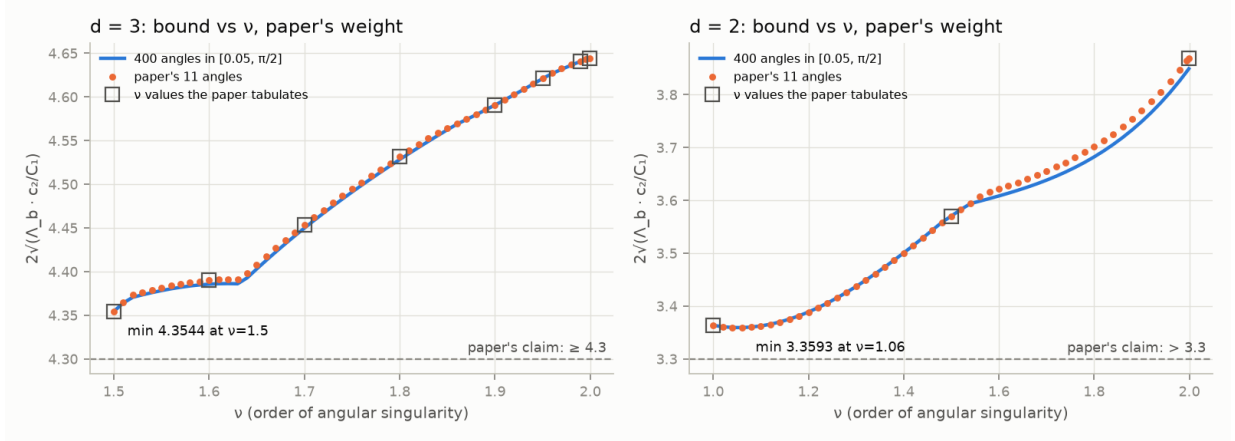


Figure 1: Bound $2\sqrt{\Lambda_b c_2 / C_1}$ against ν with the notes' weights: 400 angles (line) against the notes' eleven (dots); squares mark the tabulated ν . Dashed: the claimed thresholds 4.3 ($d = 3$) and 3.3 ($d = 2$).

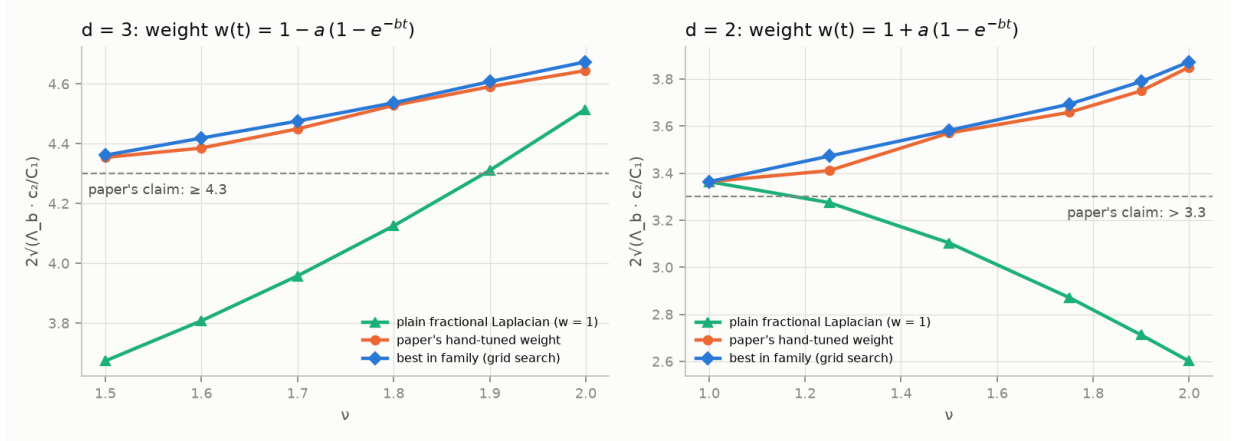


Figure 2: Plain fractional Laplacian, the notes' weight, and the best weight in its two-parameter family.

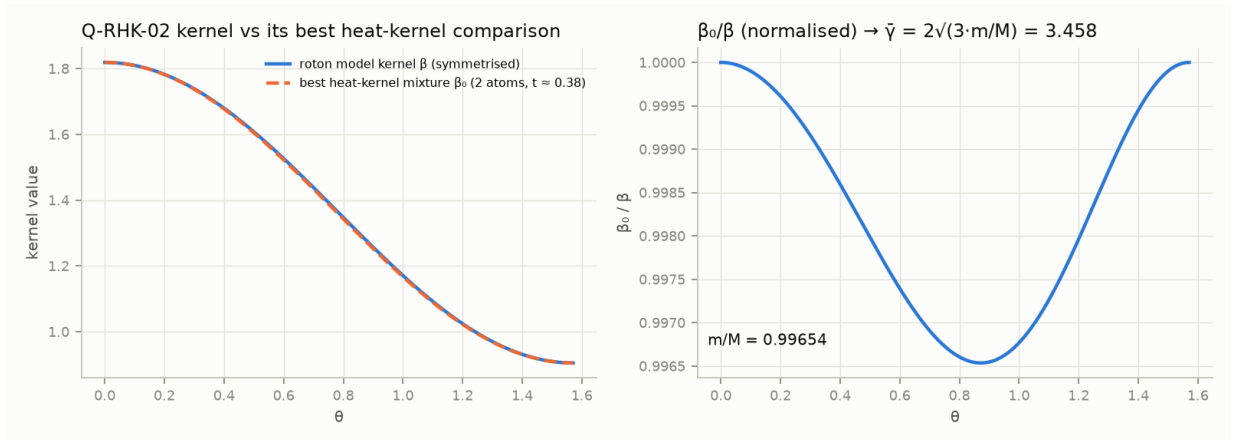


Figure 3: The Q-RHK-02 model kernel and its best heat-kernel comparison (left); their ratio, $m/M = 0.99654$ (right).

6 Machine-checked kinematics on neutron data

The companion project formalises kinematic identities used in the analysis of neutron scattering on superfluid ^4He [5, 2]. We test them on the dispersion curve Godfrin *et al.* published with [5]: ILL IN5 data at seven pressures from 0 to 24.08 bar, with per-point uncertainties, released as arXiv ancillary files. The files are fetched and checked by SHA-256, and are not redistributed. The fit reproduces the published sound velocity (1.5685 against 1.5679 meV Å) and roton gap (0.7417 against 0.7445 meV).

- *Maxon versus the roton-pair threshold.* The maxon lies below 2Δ up to 10 bar (-0.119 ± 0.001 meV) and above it at 24.08 bar ($+0.0106 \pm 0.0005$ meV, about 21σ), as reported in [6]. A proved lemma states that every time-of-flight energy scales with the incident energy, so no proportional recalibration can reverse this comparison.
- *Top of the measured range.* At saturated vapour pressure the tabulated energy at 3.60 Å^{-1} (1.554 meV) exceeds $2\Delta = 1.483$ meV, as the formalisation’s documentation asserts.
- *Three-phonon decay.* It is open if and only if the small- k dispersion curves upwards (a proved equivalence). At saturated vapour pressure a fit on $Q \leq 0.15 \text{ Å}^{-1}$ gives upward curvature at 3.8σ , consistent with the published series. The sign change reported near 18–20 bar cannot be decided from these files, which start at $Q = 0.15 \text{ Å}^{-1}$ for $P > 0$. Our first version of this test asserted it anyway; the test was redesigned and the change recorded.

7 Verification and validation

We distinguish three levels of evidence and label every claim in this paper by which level supports it.

Level 1: exact symbolic computation. The kernel coefficients, Padé approximants, dispersion polynomials, roots and minimal polynomials are exact over $\mathbb{Q}$ or $\overline{\mathbb{Q}}$. Forty-nine tests cover these and the analytic anchors of Sections 4.5–5, including the two adversarial ones described in Section 2: the kernel is re-derived from its closed form rather than compared with a table, and float input is required to raise.

Level 2: independent numerical validation. `verification/validate_zero_sound.py` checks the exact roots against 60-digit root-finding and against the asymptotic law (13), and asserts the proved bracket (14) at every reference root. It asserts the strong-coupling cases and reports the others, exiting non-zero only if a regime it claims to have solved degrades. Further scripts, all run in continuous integration, check the following:

- the real- ^3He analysis of Section 4.5;
- the neutron-data tests of Section 6;
- the reproduction of Section 5;
- Fisher-information and entropy monotonicity on the exact Bobylev–Krook–Wu solution [9] of the homogeneous Boltzmann equation for Maxwell molecules. Mass is conserved exactly, the Maxwellian limit $I = 3$ is recovered exactly, and both functionals decrease monotonically.

Level 3: proof assistant. The Lean 4 library `AgoraPhysics.Protocols` compiles against `mathlib` at the pinned revision `ec6e08b4`, and every theorem is accompanied by `#print axioms`. The scope of that evidence is worth stating plainly. `Protocols.lean` certifies identities such as the $[1/1]$ order condition $a_2 + q_1 a_1 = 0$ and the fact that $u = 10/37$ is a root of $Q - \frac{93}{10}P$. `ZeroSoundBracket.lean` and `PhaseMixingLink.lean` certify real-analysis statements about the Landau zero-sound relation and about free transport. None of them certifies anything about real ^3He or ^4He : that connection is made, where it is made, by the data-driven checks of Level 2. Two theorems in the file are labelled vacuous in the source itself: one proves $2 \cdot 2 = 4$ after unfolding a definition, and one has its hypothesis as its conclusion. They are retained, labelled, as a record.

8 Limitations, negative results and retractions

8.1 A capability the method does not have

The central negative result of this paper concerns the weak-coupling regime. The kernel (1) has a logarithmic branch point at $s = 1$, that is at $u = 1$, which is the edge of the particle-hole continuum. A Padé approximant constructed from the expansion about $u = 0$ represents this branch point by a finite set of poles, and for small F_0^s the dispersion polynomial (9) then has no root in $(0, 1)$ at all: at $M = 1$, Eq. (11) gives $u > 1$ as soon as $F_0^s < 6/5$. Higher orders do not repair this, as Table 2 shows for $F_0^s \in \{1/10, 1/2\}$.

The consequence is a retraction of a capability. The protocol registry of this project previously described tracking “the velocity ratio drop $v_{zs} \leq v_F$ into the Landau Damping continuum” by rational Padé approximants. That is not attainable by this construction, and no implementation of it ever existed in the repository. The physically correct statement is that the mode approaches the continuum edge exponentially, Eq. (13), and is exponentially inaccessible to a finite-order rational approximant built at the origin. The engine now reports `root_found: false` with a reason in these cases, and a test asserts that it continues to do so, precisely so that a future change cannot quietly return a spurious root.

8.2 Withdrawn claims of earlier versions

For the record, and because several of these claims are still reachable in the repository’s history and in earlier documents:

1. **The QVE-02 “echo”** is the Taylor series of $\text{Si}(t)^2/2$, not a plasma echo (Section 4.6).
2. **A stored artefact labelled “True 1D1V Vlasov-Poisson phase space simulation”**, with floating-point fields, was never produced by any code in this repository. No such solver exists here.
3. $L_* = 4$ is definitional (Section 4).
4. **The “Theorem 22.6” attribution** has been checked against the source and found mismatched: the paper and theorem are real, but its bound is a different, multiplicative object, not the additive formula the code computes.
5. **Three mutually inconsistent coefficient tables** once coexisted for the “Lindhard” kernel: $1/3, 19/45$ in Lean, $2/3, 2/15$ in Python, and $1/(2k + 1)$ in the prose. The resolution is that $1/(2k + 1)$ is correct for the zero-sound kernel (3), the $2/(4k^2 - 1)$ sequence belongs to the different function Eq. (4), and $19/45$ corresponded to neither and has been removed.
6. **“No open dataset exists”** for the roton regime was an earlier conclusion of this project’s data search. It was wrong: the dispersion data of [5] were published openly as arXiv ancillary files, and the search had not looked there.

7. **An automated audit of this repository** reported several of these repairs as complete before they were. In particular it stated that the coefficient tables had been unified across Python and Lean while the Lean file was untouched, and that the RPA integration was complete while the code was passing a polynomial into a routine expecting a list of Taylor coefficients, so that the “poles” it produced carried no information. Both are recorded in `RETRACTIONS.md`.

Item 7 is the reason this section exists in this form. An automated pass that writes its own audit report is capable of producing a confident, well-formatted account of work it did not do, which is the first failure mode of Section 1 applied to the act of auditing. The defence is not more prose; it is the executable checks of Section 7, which fail when a claim is false.

8.3 Other limitations

Contact with measurement is indirect and limited. The data are author-processed tables, not raw counts; raw ILL data remain to be requested. The angular kernel of Q-RHK-02 is a model, not a fit, and no available dataset constrains it. The real-parameter zero-sound analysis uses two Landau parameters and neglects F_2^s and higher, finite temperature and collisions. Its absolute velocities at the lowest pressures inherit the uncertainty of the source compressibility. The $d = 4$ evidence of Section 5 is numerical and conditional on Theorem 22.6 holding as stated.

9 Reproducibility

All results are produced by the following commands from the repository root. Python dependencies are pinned in `requirements.txt` (sympy, mpmath, numpy, scipy, matplotlib, pytest); poppler-utils provides pdftotext.

```

pip install -r requirements.txt
python3 simulations/qv_01_zero_sound.py          # exact QV-01 derivation
python3 -m pytest tests/ -q                      # 49 tests
python3 verification/validate_zero_sound.py      # Tables 2-3, bracket (3.4)
python3 verification/he3_landau/extract_table_ix.py # real 3He parameters
python3 verification/he3_landau/zero_sound_real_he3.py # Table 4
python3 verification/helium_kinematics_data.py   # Section 5
python3 verification/theorem_22_6/reproduce.py     # Section 4 (Python)
(cd verification/theorem_22_6/rust && cargo run --release) # (Rust)
verification/theorem_22_6/run.sh                 # (authors' Julia code)
cd lean4_formalization && lake exe cache get && lake build # proofs

```

The experiments of Section 5, with their run times, are listed in `verification/theorem_22_6/EXPERIMENTS.md`; `verification/THEORY_EXPERIMENT_LINKS.md` maps each theorem to the experiment that tests it.

The Lean toolchain is `leanprover/lean4:v4.31.0` with mathlib pinned at `ec6e08b4dfb4-14460f0ee7b4ce9678e7e04fb584`. Note that `lake exe cache get` is required: compiling mathlib from source takes hours, and the build that produced the evidence reported here initially failed because an interrupted clone left the dependency incomplete.

10 Conclusion

An exact-rational pipeline can deliver collective-mode velocities as algebraic numbers with proof-assistant-ready certificates, and at realistic Fermi-liquid coupling it does so accurately: 6.8×10^{-10} relative agreement with the transcendental dispersion relation at $F_0^s = 93/10$, improving

geometrically with approximant order. The same analysis shows the method has a hard boundary: the weak-coupling mode sits exponentially close to a branch point and is unreachable by rational approximants built at the origin, which retires a capability this project previously advertised.

Contact with real parameters sharpened this picture rather than confirming it. The one-parameter model, accurate as mathematics, is wrong as physics for ^3He : it puts zero sound below first sound, and the second Landau parameter restores the observed ordering, still within the exact framework. The threshold law is now a theorem, and the numerics show it is sharp. Re-examining the numerics behind a published theorem found them sound and slightly conservative. It also produced evidence that the one case the author left open is covered, which is offered to the author to check.

The methodological lesson is narrower than the physics but likely more durable. Every claim in this paper that survived did so because something executable would fail if it were false: a kernel re-derived from its closed form rather than compared against a table, a float parameter that raises instead of being silently rationalised, a validation gate that asserts only the regime it can actually solve, and an axiom audit rather than a grep for `sorry`. The claims that did not survive, catalogued in Section 8, were uniformly the ones supported by prose alone, including prose written by an automated audit about its own work.

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